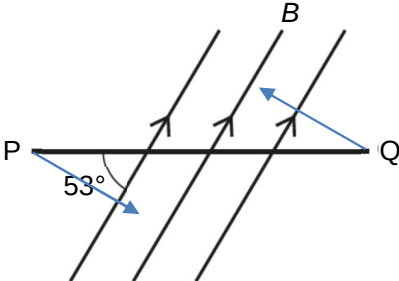
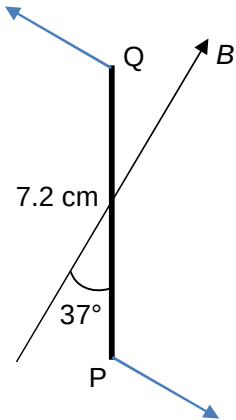
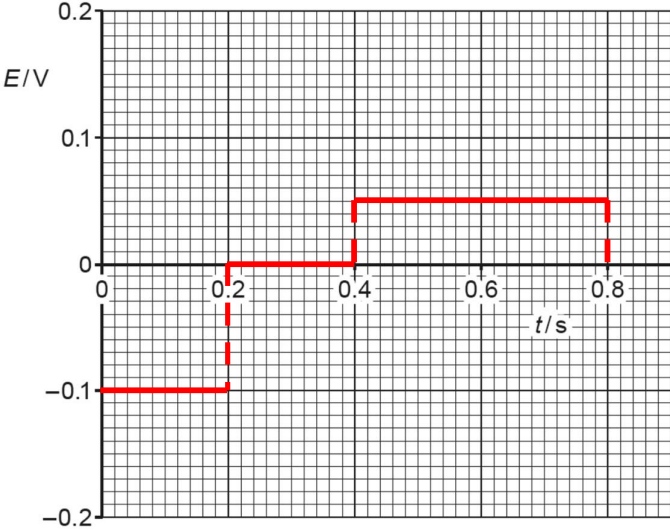


10(a)(i)	$6.0 \times 10^{-3} = B(4.2)(7.2 \times 10^{-2}) \sin 53^\circ$ $B = 0.0248 \text{ T}$ $B \approx 0.025 \text{ T}$	[1] for correct substitution [1] for correct answer
10(a)(ii)	The change in reading is a decrease. The magnetic force acting on SR is <u>upwards</u> in Fig. 10.1. OR The magnetic force acting on SR is <u>out of the plane of the paper</u> in Fig. 10.2.	[1] [1]

10(a)(iii) 1		[1] for correct direction of arrow drawn at P, perpendicular to B [1] for correct direction of arrow drawn at Q, (perpendicular to B)
10(a)(iii) 2	 Magnitude of torque = magnetic force $\times$ perpendicular distance $= [(0.025)(4.2)(8.7 \times 10^{-2}) \sin 90^\circ] [(7.2 \times 10^{-2}) \cos 37^\circ]$ $= 5.253 \times 10^{-4} \text{ N m}$ $\approx 5.3 \times 10^{-4} \text{ N m}$	[1] for correct substitution for magnetic force [1] for correct substitution for perpendicular distance [1] for correct answer
10(a)(iii) 3	The frame will rotate until it is <u>perpendicular to the magnetic field.</u>	[1]
10(b)(i)	Magnetic flux linkage in smaller solenoid $= (320)1.4 \times 10^{-3} B$ $= 0.448B$ Magnetic flux linkage in larger solenoid $= (40)4.2 \times 10^{-3} B$ $= 0.168B$ The <u>smaller solenoid</u> has a <u>larger magnetic flux linkage.</u>	[1] for correct substitution [1] correct conclusion

10(b)(ii) 1	Magnitude of e.m.f. induced $= (320) \left(1.4 \times 10^{-3} \right) \left(\frac{45 \times 10^{-3}}{0.20} \right)$ $= 0.1008 \text{ V}$ $\approx 0.10 \text{ V}$	[1] for correct substitution [1] for correct answer
10(b)(ii) 2		[1] for horizontal line at -0.10 V from 0 to 0.20 s [1] for horizontal line at zero from 0.20 s to 0.40 s [1] for horizontal line at 0.05 V from 0.40 s to 0.80 s
10(b)(iii)	As the smaller solenoid is removed, the <u>flux linkage</u> in the smaller solenoid is <u>decreasing</u>. According to Faraday's Law of electromagnetic induction, this causes an <u>induced e.m.f.</u> in the smaller solenoid. The induced e.m.f. causes an <u>induced current</u> to flow in the smaller solenoid. The induced current causes an <u>induced magnetic field</u> around the smaller solenoid according to Lenz's Law. The two magnetic fields interact to create an <u>attractive force</u> between the solenoids to oppose the change which is the smaller coil being removed.	[1] [1] [1]